Machine Learning II - Structured Models

Exercise sheet **Min-Cut Based Inference**

Bogdan Savchynskyy - Bogdan.Savchynskyy@tu-dresden.de

Exercise 1.

Algorithms. Compare time and attained energy/bound for the following algorithms:

- α-expansion;
- $\alpha\beta$-swap;
- α-expansion fusion;
- `TRW-S`.

Command lines. Use the parameter `-o` to specify a file for the inference result, e.g.

```
-o pfau-small.alpha-exp.h5
```

and the `HDF` format viewer to check the timing and corresponding value/bound:

```
hdfview  pfau-small.alpha-exp.h5 &
```

For α-expansion compare the native `OpenGM` implementation

```
opengm_min_sum -a ALPHAEXPANSION -m dataset-name.h5 -p 1 -v -o dataset-name.ogm-alpha-exp.h5
```

to the implementation from the `MRF` Library:

```
opengm_min_sum -a MRF --inference EXPANSION -m dataset-name.h5 -p 1 -v -o dataset-name.mrf-alpha-exp.h5
```

For $\alpha\beta$-swap use

```
opengm_min_sum -a MRF --inference SWAP -m dataset-name.h5 -p 1 -v -o dataset-name.mrf-swap.h5
```

Parameters for α-expansion fusion read:

```
opengm_min_sum -a ALPHAEXPANSIONFUSION -m dataset-name.h5 -p 1 -v -o dataset-name.alpha-exp-fusion.h5
```

Run `TRW-S` as many iterations as needed to attain the optimal value of all other algorithms, but not more than 1000 iterations.
Use parameters `--energy TL1` for (truncated) ℓ_1 pairwise factors, `--energy TL2` for (truncated) ℓ_2 and `--energy TABLES` for all other. Use `--minDualChange 0` to prevent early stopping of the algorithm. Command line example:

```
opengm_min_sum -a TRWS --energy TABLES --maxit 1000 -m dataset-name.h5 -p 1 -v -o dataset-name.trws.h5
```

Datasets. Use the following datasets for comparison:

1. `color-seg-n4/pfau-small.h5` (Potts)
2. `mrf-stereo/ted.h5` (Potts)
3. `mrf-stereo/ven.h5` (TL2)
4. `mrf-photomontage/pano-gm.h5` (semi-metric)
5. `mrf-inpainting/penguin.h5` (TL2)